

Chemical Coordination & Integration

SHORT NOTES

Source	Hormone	Target	Action
1. Hypothalamus	Releasing and inhibiting hormones	Pituitary	Controls release of tropic hormones by pituitary.
	ADH and Oxytocin	Sends them for storage in posterior pituitary for later release.	ADH regulates amount of water while oxytocin causes contraction of uterus and milk ejection.
2. (a) Anterior Pituitary (tropic hormones)	TSH (Thyroid stimulating hormone)	Thyroid	Secretion of T_4 and T_3 .
	ACTH (Adrenocorticotrophic hormone)	Adrenal cortex	Secretion of glucocorticoids.
	FSH (Follicle stimulating hormone) and LH (Luteinizing hormone)	Ovaries and testes	Regulates oogenesis and spermatogenesis.
(b) Anterior Pituitary (hormones)	PRL (Prolactin)	Mammary glands	Production of milk.
	GH (Growth hormone)	Bones and muscles	Stimulates growth.
3. Pineal	Melatonin	Body	Circadian rhythms and pigmentation.
4. Thyroid	T_4 (Thyroxine) and T_3 (Triiodothyronine)	General body	Increases cellular metabolism.
	Thyroid calcitonin	Bone	Lowers blood Ca^{2+} .
5. Parathyroid	PTH (Parathyroid hormone)	Bone	Increases blood Ca^{2+}
6. Thymus	Thymosin	Immunity	It helps in maturation of T-lymphocyte, resistance to infection and attainment of sexual maturity.
7. Pancreas (alpha cells)	Glucagon	Liver, muscles, fat	Increases blood glucose.
8. Pancreas (beta cells)	Insulin	Liver, muscles, fat	Lowers blood glucose.
9. Pancreas (PP-cells)	Somatostatin	Pancreas	Inhibits release of insulin and glucagon.
10. Adrenal gland (cortex)	Glucocorticoids (e.g., cortisol) and mineralocorticoids (e.g., aldosterone)	General body and kidney	Increases blood glucose, increases reabsorption of Na^+ and excretion of K^+ .
	Sex corticoids	Body	Male like sexual characters.
11. Adrenal gland (medulla)	Epinephrine (adrenaline) and norepinephrine (noradrenaline)	Blood vessels, liver and heart	Increases blood glucose, constricts blood vessels (fight or flight response).
12. Testis	Testosterone	Testes and general body	Spermatogenesis, secondary sexual characteristics.
13. Ovary	Estrogen and progesterone	Ovary, uterus and general body	Menstrual cycle, secondary sexual characteristics and pregnancy.

Mechanism of Hormone Action

- ❖ Hormones could be peptides (insulin, glucagon), steroids (testosterone), iodothyronines (thyroxine) and amino acid derivatives (epinephrine).
- ❖ Some hormones bind to extracellular receptors and generate secondary messengers which in-turn regulate cellular metabolism. Other hormones (steroid hormones) interact with intracellular receptors and bring about changes by influencing gene expression.